

A BEGINNERS' GUIDE TO ATMEL AVR DEVELOPMENT



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Learning microcontrollers is difficult, as any book meant for them starts with dry theory of their architecture and jargon of Assembly language programming. Most of my students frequently come to me ask-

ing “where to start from” and “how to proceed.” I always suggest them a line of readings, proper tools and software which one should learn to begin with a microcontroller. Here is a guide to Atmel AVR development which would be very helpful to beginners for learning microcontrollers.

tools for embedded systems. It can be downloaded for free from IAR website www.iar.com. IAR Embedded Workbench is being parallelly developed by IAR Systems and ATMEL developers and hence it generates the optimised code which uses full ‘C’ coding capabilities of AVR devices. Coding in IAR

is nicely supported by its library functions, which makes a normal C user comfortably program any embedded device too.

Let us begin using our coding platform. To start the application, click ‘IAR Embedded Workbench’ icon in ‘All Programs’

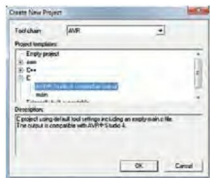


Fig. 1: Creating new project window

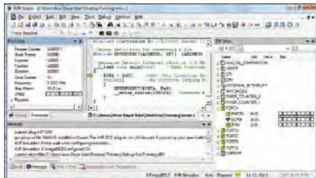


Fig. 4: AVR Studio window for simulation

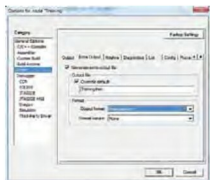


Fig. 2: Options window for selection of Intel-standard hex file



Fig. 3: Main window for compilation of source program

Any microcontroller would typically have some input/output pins, a few of which may have multiple functions. It has a reset switch, DC supply pins and a clock source which helps the device to send the data through different sections within the chip.

Without going much into the details of a microcontroller, let us learn practically by implementing various parts of a microcontroller. For this reason, I would like to introduce ATmega8515, which belongs to the 8-bit family of AVR microcontrollers—one of the most basic ones.

All AVR microcontrollers are similar when it comes to programming and implementation, apart from some of the added features. The only prerequisite for a novice learner is knowledge of binary arithmetic and C programming basics.

IAR Embedded Workbench

The IAR Embedded Workbench provides a suite of AVR development

menu of Windows. From the main menu, select Project→Create New Project and start an AVR Studio compatible C project (refer Fig. 1), as later on you will need to simulate the code in AVR Studio 4.0. Save the new project to a new folder as an embedded workbench project (.ewp) file and then press ‘Alt+F7.’ ‘Options’ window appears as shown in Fig. 2. Select the Atmega8515 microcontroller as follows:

Project→Options→General Options (under Category)→Target→Processor Configuration→Select cpu=m8515, Atmega8515

Enable bit definitions in I/O include files as follows:

Project→Options→General Options (under Category)→System→Enable bit definitions in I/O include files

Generate the Intel hex file for burning the code into ATmega8515 microcontroller as follows (refer Fig. 2):

Project→Options→Linker (under Category)→Extra Output→Generate